

Article

Solutions for Modelling the Marine Oil Spill Drift

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Abstract: Oil spills represent a critical environmental hazard with far-reaching ecological and economic consequences, necessitating the development of sophisticated modelling approaches to predict, monitor, and mitigate their impacts. This study presents a computationally efficient and physically grounded modelling framework for simulating oil spill drift in marine environments, developed using Python coding. The proposed model integrates core physical processes—advection, diffusion, and degradation—within a simplified partial differential equation system, employing an integrator for numerical simulation. Building on recent advances in marine pollution modelling, the study incorporates real-time oceanographic data, satellite-based remote sensing, and subsurface dispersion dynamics into an enriched version of the simulation. The research is structured in two phases: (1) the development of a minimalist Python model to validate fundamental oil transport behaviours, and (2) the implementation of a comprehensive, multi-layered simulation that includes NOAA ocean currents, 3D vertical mixing, and support for inland and chemical spill modelling. The results confirm the model’s ability to reproduce realistic oil spill trajectories, diffusion patterns, and biodegradation effects under variable environmental conditions. The proposed framework demonstrates strong potential for real-time decision support in oil spill response, coastal protection, and environmental policy-making. This paperwork contributes to the field by bridging theoretical modelling with practical response needs, offering a scalable and adaptable tool for marine pollution forecasting. Future extensions may incorporate deep learning algorithms and high-resolution sensor data to further enhance predictive accuracy and operational readiness.



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1. Introduction

Oil spills represent a critical environmental hazard with far-reaching ecological and economic consequences, necessitating the development of sophisticated modelling approaches to predict, monitor, and mitigate their impacts. The modelling of oil spill drift involves simulating the movement of spilled oil influenced by oceanographic, meteorological, and chemical processes. Various models have been developed to improve prediction accuracy, leveraging advancements in remote sensing, hydrodynamic modelling, and machine learning techniques.

The major goal of this study is to develop a computationally simple, but physically realistic model based on Python algorithm, to simulate oil spill dynamics in marine environments, providing insights into their evolution and response mechanisms, starting from the state-of-the-art concepts and studies pursued in this domain. The mathematical

modelling of oil spill drift is essential for understanding how hydrocarbons behave in aquatic environments and for designing effective mitigation strategies.

Considering the fundamental approaches to oil spill drift modelling the oil spill drift modelling is typically based on Lagrangian particle tracking models and Eulerian grid-based models. These models incorporate factors such as wind forcing, ocean currents, Stokes drift, and wave interactions. In support of hydrocarbon drift forecasting the assimilation of satellite observations present a satellite data assimilation framework for oil spill modelling, several studies evaluating various global oil spill models, highlighting the effectiveness of satellite observations in improving forecast accuracy [1]. Eco sensitivity as an operational model for oil spill management according to some authors introduces the Accidental Damage and Spill Assessment Model (ADSAM), integrating ecological sensitivity into oil drift modelling, the existing studies emphasizing the importance of ecological parameters in oil spill response and preparedness [2].

Approaching the advanced modelling techniques, recent studies have explored numerical modelling, machine learning, and probabilistic forecasting to enhance oil spill drift predictions. Therefore, Deepwater Oil Spill Model (DWOSM) and Stochastic Modelling are employing a deepwater oil spill model (DWOSM), combined with a stochastic approach destined to assess the risk of hydrocarbon dispersion in deep-sea environments. In this perspective, the conducted studies have revealed that surface oil drifts predominantly towards the southeast and east, highlighting regional drift patterns [3]. Moreover, the influence of Stokes Drift on Oil Spill Simulation investigates the role of Stokes drift in oil spill movement using the FVCOM-SWAVE model, the recent findings demonstrating that incorporating Stokes drift significantly alters the trajectory of oil spills, providing better agreement with satellite Synthetic Aperture Radar (SAR) images [4].

As most recent trend, machine learning and remote sensing for oil spill prediction have been developed, AI and remote sensing technologies being further integrated into oil spill modelling to improve accuracy and efficiency, supported in practice by following often adopted solutions:

1. adversarial neural networks for oil spill prediction have proposed an adversarial temporal convolutional network for correcting forecasted sea surface dynamic fields, significantly enhancing oil spill drift predictions [5];
2. support vector regression for wind drift parameterization alternative has developed a support vector regression (SVR) model for wind drift parameterization, improving oil spill drift simulations in the Red Sea [6].

In practice, several studies for real-life scenarios have been conducted in research applying drift models to real spill scenarios to assess their effectiveness, the most known being the following ones:

1. satellite-based oil spill emergency response model highlighted the integration of satellite remote sensing data with oil drift modelling to enhance emergency response and recovery efforts [7];
2. shoreline response and traditional fishing communities have assessed oil spill drift impacts on traditional fishing communities using worst-case scenario drift modelling, identifying coastal impact zones and improving training for response teams [8].

In support of authors' study hypothesis definition, the literature review has provided a comprehensive overview of the latest developments in oil spill drift modelling, addressing both technical advancements and real-world applications, the listing references from Table 1 summarizing the actual key concepts in oil spill drift modelling.

Table 1. Oil Spill Drift Modelling Concepts.

Concept	Description	Reference
Lagrangian and Eulerian Models	<i>Lagrangian models track oil particles, while Eulerian models use grid-based computations to simulate oil movement.</i>	[1]
Accidental Damage and Spill Assessment Model (ADSAM)	<i>A new oil spill drift model integrating ecological sensitivity and damage assessment.</i>	[2]
Stochastic Modeling of Oil Spills	<i>Uses probability-based techniques to predict oil spill drift patterns.</i>	[3]
Stokes Drift and Oil Transport	<i>Investigates the role of wave-induced motion in oil drift predictions.</i>	[4]
Adversarial Neural Networks for Oil Spill Prediction	<i>AI-based approach improving oil spill drift forecasting by correcting ocean current forecasts.</i>	[5]
Support Vector Regression (SVR) for Oil Spill Modelling	<i>Machine learning technique applied to parameterize wind-driven drift of oil spills.</i>	[6]
Satellite Remote Sensing in Oil Spill Detection	<i>Uses satellite imagery to track and model oil spill movement.</i>	[7]
Shoreline Impact Assessment and Oil Drift Models	<i>Examines oil spill drift towards the coastline using worst-case scenario models.</i>	[8,9]
Deepwater Oil Spill Modelling (DWOSM)	<i>A model designed for deep-sea oil spill drift and subsurface hydrocarbon transport.</i>	[3,10]
Integration of Multi-Source Observational Data	<i>Combines satellite, drone, and radar data for real-time oil spill monitoring.</i>	[5,11]

(Source: authors' collection from literature review).

Although the recent progresses in scientific research are consistent and relevant, while oil spill drift modelling has advanced significantly, challenges are still remaining especially regarding the model uncertainty, real-time data assimilation, and computational efficiency, future research providing a significant potential for following focus approaches:

- integrating multi-source observational data (e.g., drones, radar, satellite images) for real-time monitoring;
- developing AI-driven models capable of learning from historical oil spill events;
- improving prediction accuracy in complex hydrodynamic environments.

Oil spill drift modelling will continue to play a crucial role especially in case of marine pollution management and emergency response, involving in deeper manner modern models leveraging satellite data, AI, and numerical simulations to improve prediction accuracy [12]. However, continued advancements are required to enhance real-time forecasting capabilities and reduce model uncertainty. For this purpose, the authors of present research paper are aiming to identify a reliable alternative for modelling marine oils spills drift, starting from the most recent applied models identified in the literature, with the valorisation of Python 3.13.0 software capabilities.

Following the conclusions from literature review, for a coherent approach the authors have followed the next logical workflow for research:

- problem identification by disclosing the environmental challenges posed by marine oil spills, to highlight the need for accurate, real-time modelling tools;
- analysis of identified models of marine oils spills drift to disclose the applied parameters in oil spill drift's simulation practices;
- the development of a minimalist Python simulation and the implementation of a 2D RK4-integrated Python simulation of oil drift, validating the physical behaviour with basic parameters;

- the expansion of model functionality by integration of NOAA ocean currents, vertical dispersion, chemical simulation, and remote sensing to enrich the model with real-world data and 3D subsurface dynamics;
- model validation in reference to real-world applications, including Black Sea simulations and worst-case scenarios, to confirm the model flexibility and cross-domain utility;
- results analysis and interpretation of simulations and model behaviour under varied conditions;
- analysis of potential future research directions, highlighting the innovative aspects and suggest future improvements.

To position the proposed model developed in the Section 4 within the broader landscape of marine drift modelling solutions, it is important to highlight its unique contributions compared to existing open-source platforms such as GNOME (General NOAA Operational Modeling Environment), OpenDrift, and Parcels. These widely adopted frameworks focus primarily on Lagrangian particle tracking, offering robust simulation capabilities for oil spills and drifting materials using real-time oceanographic inputs and dynamic environmental forcings. However, their inherent complexity, higher dependency on external libraries, and black-box architecture may limit flexibility for rapid prototyping or interdisciplinary adaptation. In contrast, the Python-based model developed in this study follows a simplified Eulerian advection–diffusion–reaction formulation, emphasizing clarity, transparency, and didactic value. It integrates core physical processes—such as surface drift, lateral spreading, evaporation, and biodegradation—within a customizable finite-difference framework. Moreover, the model incorporates a novel remote sensing feature using OpenCV, enabling the detection and initialization of oil spills directly from satellite or UAV imagery. This functionality, combined with the inclusion of NOAA ocean current data, 3D vertical mixing, and multi-scenario simulation potential (e.g., inland chemical spills or drifting sea mines), renders the framework highly versatile. Thus, this contribution offers a valuable complement to existing operational models—bridging academic modeling and practical deployment—while maintaining computational accessibility and cross-domain adaptability for researchers, educators, and environmental analysts.

2. Problem Identification and Analysis of Marine Oils Spills Drift Modelling Solutions

Modern oil spill modelling solutions have combined numerical simulations, AI, and remote sensing to improve accuracy and efficiency. Continued advancements will further enable faster and more effective responses to oil spills, minimizing environmental and economic damage, in which context the authors have aimed develop a dedicated algorithm, valuing the Python 3.13.0 software capability on dynamic modelling.

Oil spill modelling involves simulating the transport, dispersion, and fate of oil in marine and coastal environments, these models enhancing the prediction capability of the oil spills movement, aiding in the emergency response and the environmental impact assessments. Modern oil spill modelling solutions combine numerical simulations with advanced data-driven techniques for improved accuracy.

As prerequisite for present research, the authors have searched to identify applied examples of modelling solutions, offered by different authors in recent researching, who followed different algorithms, in order to formulate further suggestions for algorithm design, depicted as following:

- (a) Numerical investigation of dominant factors in the diffusion behaviour of oil slicks on the water surface—the proposed model has used the advection-diffusion equation method to simulate oil slick dispersion, highlighting the importance of selecting the right diffusion models for large-scale oil spill incidents [12]. This research emphasizes

how environmental conditions (wind, temperature, and water currents) impact oil dispersion, validating the present approach of the authors of using advection-diffusion modelling to simulate oil spills.

- (b) Modelling the fate and transport of pollutants in the marine environment—another identified study has approached the models of oil spills, ship emissions, and offshore pollution transport, integrating the advection, diffusion, and sedimentation effects for a more comprehensive simulation [13]. The authors have analysed high-resolution modelling techniques to improve oil spill predictions, suggesting to the authors of present research the extension of the modelling, incorporating pollution emissions from shipping and offshore drilling and including the sedimentation effects and high-resolution environmental interactions.
- (c) Modelling and forecasting of mesoscale circulation and oil pollution transport at sea—the study aimed to model the oil pollution transport on example of Black Sea region, using a non-stationary advection-diffusion equation, incorporating mesoscale ocean circulation and environmental factors, validated with real ocean current data to improve forecasting accuracy [14]. The model has suggested to the authors to enhance the accuracy of oil spill simulations by incorporating real-world ocean circulation data in the model algorithm.
- (d) Numerical study of subsurface oil spills—the study has been focused on subsurface oil spills, unlike many studies that only consider surface spills, using a Lagrangian framework namely OceanParcels, a set of Python classes and methods to create customisable particle tracking simulations using output from Ocean Circulation models (Parcels: Probably A Really Computationally Efficient Lagrangian Simulator, available on: <https://oceanparcels.org> (accessed on 1 January 2025)), to model oil dispersion with an advection-diffusion equation. This model incorporates vertical mixing processes in oil transport [15,16] that suggested to the authors the expansion of proposed algorithm to simulate also the subsurface oil spills, by incorporating vertical advection-diffusion dynamics.
- (e) Numerical Investigation of Dominant Factors in the Diffusion Behaviour of Oil Slicks on the Water Surface—uses the advection-diffusion equation to model oil slick behaviour, emphasizing the model selection and parameter tuning for more accurate large-scale oil spill simulations [12]. The relevance of this version of modelling revealed to the authors the necessity of refining the parameter's selection to ensure more accurate diffusion and advection modelling.

Based on the literature review synthesized in Table 2 outlines, the authors have concluded that to enhance the oil spill simulation, the design model and its related algorithm should:

- consider implementing more dynamic environmental conditions, integrating the ocean currents (e.g., variable wind speeds and ocean currents), based on the usage of real-time ocean current data (e.g., NOAA datasets, available on: <https://www.noaa.gov> (accessed on 1 January 2025)) to improve accuracy [17];
- consider the sedimentation and biodegradation effects that would make the model even more realistic;
- include remote sensing & autonomous tracking—by adding drones or satellite imagery processing for oil spill detection;
- simulate subsurface oil spills—the model should include vertical mixing and sinking effects;
- comprise the inland water pollution simulation;
- optimize the parameters and & validate the model, by comparing the results with real-world oil spill events for accuracy;

- include the chemical transport modelling, by adapting the code to simulate toxic chemical spills, not just oil.

Table 2. Research findings related to oil spill simulation case studies using diffusion and advection modelling.

Study	Key Focus in Modelling	Modelling Findings	Relevance to Proposed Modelled Algorithm	Author
Numerical Investigation of Dominant Factors in the Diffusion Behaviour of Oil Slicks	Parameter selection for diffusion modelling	<i>Analyses how diffusion coefficients affect oil spread accuracy</i>	Fine-tune diffusion and advection parameters based on real spill events	[12]
Modelling the Fate and Transport of Pollutants in the Marine Environment	Chemical pollution modelling	<i>Extends oil spill modelling to chemical dispersion</i>	Expand the model to simulate toxic chemical spills, not just oil	[13] [18]
Modelling and Forecasting of Mesoscale Circulation and Oil Pollution Transport in the Black Sea	3D advection-diffusion model	<i>Incorporates real-world ocean currents to enhance oil spill forecasting</i>	Integrate real-time ocean circulation data for better accuracy	[14]
Multi-USV Cooperative Oil Spill Source Seeking	Remote sensing and oil tracking using autonomous vehicles	<i>Uses Unmanned Surface Vehicles (USVs) and diffusion-advection PDEs for oil spill detection</i>	Extend the model to autonomous spill tracking using remote sensing	[19]
A Numerical Study of Subsurface Oil Spills	Subsurface oil dispersion	<i>Uses OceanParcels framework to model oil mixing in water vertically</i>	Add vertical mixing and sinking effects to simulate underwater spills	[15,16]
Study on the Diffusion Law of Floating Pollutants in Inland Waters	Inland water pollution modelling	<i>Uses particle-based advection-diffusion modelling to study oil slick behaviour in rivers and lakes</i>	Extend your model to simulate oil spills in inland water bodies	[20]

(Source: authors' collection from literature review).

3. Research Method

Reaching the present study major objective, the oil spill modelling should involve the simulation of the transport, dispersion, and fate of oil in marine and coastal environments, aiming the real-time support in predicting the movement of oil spills, aiding in emergency response and environmental impact assessments. Modern oil spill modelling solutions should combine numerical simulations with advanced data-driven techniques for improved accuracy [11].

3.1. Vertical Evolution of Hydrocarbon Films Spilled in Water

The degradation mechanism of hydrocarbons spilled in aquatic environments involves a wide range of interrelated physical, chemical, and biological processes as suggested in Figure 1. These processes include dispersion, evaporation, emulsification, photooxidation, biodegradation, and sedimentation, each playing a crucial role in determining the environmental fate of oil spills and informing appropriate response strategies [10,21].

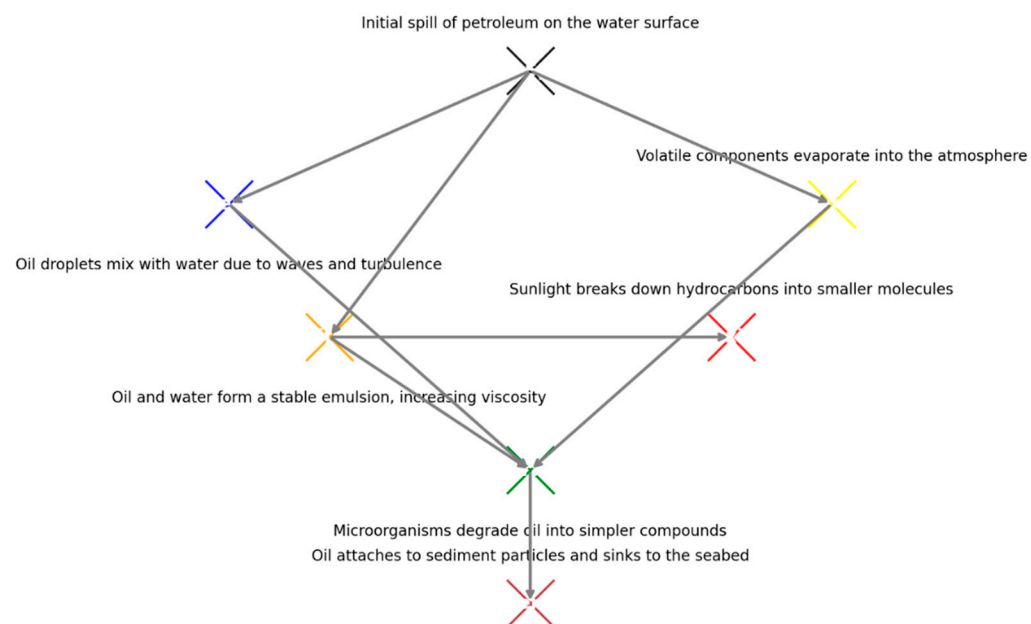


Figure 1. Mechanism of Degradation of Oil Spilled on Water Surface. (Source: authors' design following the literature review outlines).

(a) Dispersion of Hydrocarbons in Water

Dispersion refers to the incorporation of hydrocarbon droplets into the water column, a process predominantly influenced by wave action and water turbulence. When waves break, they entrain small hydrocarbon droplets into the water column; if these droplets are small enough (≤ 70 microns), the natural turbulence of the water keeps them suspended, preventing resurfacing, by analogy to how air turbulence suspends dust particles in the atmosphere. Consequently, larger droplets (> 70 microns) quickly return to the surface due to buoyancy forces [22]. The extent of natural dispersion is governed by: wave energy, which enhances mixing; volume of oil spilled per unit of water, and viscosity of the petroleum product, which determines droplet formation and stability.

The dispersion rate is directly proportional to wave turbulence and the volume of hydrocarbons released but inversely proportional to viscosity [23]. Under extreme turbulence, up to 80–90% of spilled petroleum can become dispersed, leading to cases where no technical intervention was required (e.g., North Cape, USA, and Shetland Islands, UK) [10].

Variability in hydro-meteorological conditions significantly affects dispersion efficiency. For instance, the proportion of dispersed hydrocarbons ranges between 2.5% and 12% of the total spill volume, depending on wind speed (1 m/s vs. 15 m/s), wave height (0.3 m vs. 3.5 m), and temperature (7 °C vs. 20 °C) [22].

A major consequence of dispersion is the reduction in surface oil film volume, while simultaneously increasing oil density and viscosity. These factors allow for quantitative estimations of residual oil properties, essential for designing optimal intervention techniques.

(b) Evaporation

Evaporation occurs within the first few hours of an oil spill and primarily affects volatile fractions, the rate of evaporation depending on various variables as: wind speed, air and water temperatures, surface area of the slick or composition of the hydrocarbons (volatile fraction content). Typically, petroleum products lose 30–40% of their total volume through evaporation, with higher temperatures and wind speeds accelerating the process [10]. Under calm conditions (1 m/s wind, 0.3 m waves, 7 °C water), evaporation is significantly lower than in turbulent conditions (15 m/s wind, 3.5 m waves, 20 °C water) [24]. As lighter hydrocarbons evaporate, the remaining oil becomes denser and

more viscous, reaching 990–1000 kg/m³ in density and 10,000 cSt viscosity within 24–36 h post-spill [9]. In cases of light refined petroleum products (e.g., gasoline, diesel, kerosene), evaporation may result in complete dissipation within 12–24 h, posing a significant explosion hazard due to flammable gas accumulation [10]. Additionally, evaporation releases toxic and carcinogenic compounds into the atmosphere. The U.S. National Institute for Occupational Safety and Health specifies a maximum permissible exposure limit of 1 ppm over an 8-h work shift, 40-h workweek for safety compliance [25].

(c) Emulsification

Emulsification occurs when continuous wave agitation induces water droplet entrapment within oil, forming a water-in-oil emulsion. This process is largely dependent on the presence of heavy petroleum fractions, such as asphaltenes and resins ($\geq 0.5\%$ content leads to emulsification), or oil viscosity, which influences the stability of emulsions. During evaporation, the proportion of asphaltenes and resins increases, enhancing water absorption and stabilizing the emulsion. Within 24 h, emulsification can reach 80% water content, transforming an initial 400-ton oil spill into 2000 metric tons of emulsified product [21]. The emulsification rate varies with oil viscosity in the sense that a low-viscosity oils emulsify within hours and a high-viscosity oils emulsify over several days.

The resulting “chocolate mousse” appearance (black turning to brown/orange) is well-documented in oil spill studies [22]. Thixotropic behaviour is also observed, emulsified oil remaining fluid under wave motion but becoming semi-rigid in stagnant conditions. Emulsion stability depends on temperature, with warmer climates promoting breakdown.

(d) Photooxidation

Photooxidation is a chemical reaction between hydrocarbons and oxygen, accelerated by ultraviolet (UV) radiation, this reaction breaking down hydrocarbons into smaller, more soluble, and biodegradable compounds [10]. Lighter hydrocarbons undergo oxidation rapidly, increasing water solubility and biodegradability and heavy hydrocarbons or weathered oil undergo polymerization, reducing their biodegradation potential.

(e) Biodegradation

Biodegradation is the microbial breakdown of hydrocarbons into oxidized by products, as exemplified in Table 3 based on authors’ calculations for South-Western Black Sea region. The degradation rate depends on temperature ($>25\text{ }^{\circ}\text{C}$ promotes microbial activity), oxygen availability, nutrient presence (nitrogen, phosphorus) and on hydrocarbon type (lighter hydrocarbons degrade faster). Under optimal conditions, Mediterranean marine bacteria can degrade 1 g of oil per square meter per day [9]. However, biodegradation is slow (weeks to years) and often incomplete, due to the complex hydrocarbon structures in crude oil and to the limited nutrient availability in marine environments.

Table 3. The balance of biodegradation process (Source: authors’ calculations).

1 kg HC	Will be transformed in	1.6 kg CO ₂
+2.6 kg O ₂		1 kg H ₂ O
+70 g N		1 kg biomass

(f) Sedimentation

Sedimentation occurs when oil particles adhere to suspended sediment, forming denser aggregates that sink to the seabed. The process is enhanced by turbulence (0–1000 g sediment/m³ water) being influenced by wave energy, which facilitates oil-particle aggregation. While sedimentation reduces surface oil, it is less effective than dispersion or

evaporation and understanding sediment dynamics is essential for optimizing oil spill response technologies [10].

In conclusion, the vertical evolution of hydrocarbons in aquatic environments is dictated by interdependent processes considered variable of modelling in algorithm development, respectively: dispersion, evaporation, emulsification, photooxidation, biodegradation, and sedimentation as suggested in Figure 1. These processes are highly variable, depending on environmental conditions and spill characteristics, significantly impacting response strategies and mitigation efforts.

3.2. Definition of Modelling Problem

The major goal of this study is to develop a computationally simple, but physically realistic model based on Python algorithm, to simulate oil spill dynamics in marine environments, providing insights into their evolution and response mechanisms. The mathematical modelling of oil spill drift is essential for understanding how hydrocarbons behave in aquatic environments and for designing effective mitigation strategies, the algorithm development asking for following stages:

- the formulation of a simplified marine drift model for oil spills in water, based on the fundamental physical processes of advection (transport), diffusion (spreading), and reaction (evaporation/degradation);
- the development and the implementation of this model in Python, focusing on a minimalist approach for clarity
- the Python algorithm development and enrichment, in an enlarged approached, considering all identified variables in the model, as described above.

(a) Fundamental Equations Governing Oil Spill Dynamics

Mathematical models for oil spills typically use partial differential equations (PDEs) to describe transport phenomena in aquatic environments. These equations account for three primary processes:

1. advection—the transport of oil by currents and winds;
2. diffusion—the spreading of oil due to turbulence and molecular motion;
3. reaction—the transformation of hydrocarbons through evaporation, dissolution, biodegradation, and chemical oxidation.

1. Advection Equation

Advection refers to the transport of a scalar field (oil concentration) due to fluid motion. The velocity field can either be constant or time-dependent.

For the one-dimensional (1D) case, where scalar field $f = f(t, x)$ represents the oil concentration at a point x , the advection equation is:

$$\frac{\partial f(t, x)}{\partial t} = -v \cdot \frac{\partial f(t, x)}{\partial x} \quad (1)$$

where v is the velocity of the current.

For the general three-dimensional (3D) case, the equation extends to:

$$\frac{\partial f(t, x, y, z)}{\partial t} = -v \cdot \nabla f(t, x, y, z) \quad (2)$$

where $v = (v_x, v_y, v_z)$ representing the velocity vector representing current-induced advection

The advection process primarily governs large-scale oil movement and is crucial for predicting spill trajectory [22].

(b) Diffusion Equation

Diffusion accounts for the spreading of oil due to molecular motion and oceanic turbulence, following Fick's laws of diffusion.

For the one-dimensional (1D) case, the diffusion equation is:

$$\frac{\partial f(t, x)}{\partial t} = D \cdot \frac{\partial^2 f(t, x)}{\partial x^2} \quad (3)$$

For the three-dimensional (3D) case, it generalizes to:

$$\frac{\partial f(t, x, y, z)}{\partial t} = D \cdot \nabla^2 f(t, x, y, z) \quad (4)$$

where D is the diffusion coefficient, which varies with turbulence intensity and oil properties.

Diffusion plays a significant role in determining the spatial extent of the spill and is influenced by wind-generated turbulence and water column stratification [10].

(c) Reaction-Decomposition Equation

Oil undergoes evaporation, dissolution, biodegradation, and photooxidation, all of which can be modelled as reaction kinetics.

(d) First-Order Degradation Kinetics:

If the degradation follows first-order kinetics, the governing equation is:

$$\frac{\partial f(t, x, y, z)}{\partial t} = -\text{constant} \cdot f(t, x, y, z) \quad (5)$$

This equation shows an exponential decrease of the field $f(t, x, y, z)$, where $f(t, x, y, z)$ is the reaction rate constant, leading to exponential decay in oil concentration over time.

(e) Zero-Order Degradation Kinetics:

Alternatively, if degradation follows zero-order kinetics, the equation will indicate the linear decrease in oil concentration over time, being simplified to next formula:

$$\frac{\partial f(t, x, y, z)}{\partial t} = -\text{constant} \quad (6)$$

The choice of kinetic model depends on oil type and environmental conditions [9].

3.3. Theoretical Framework for Oil Spill Drift Simulation

This simplified but scalable problem formulation forms the basis for the simulation logic developed in Section 4.1 and extended in Section 4.2. Each cell in the simulation grid represents an area of approximately X square meters, and each timestep corresponds to Y minutes of environmental evolution. Constants such as advection and diffusion rates were selected based on values commonly reported in oil spill literature [26].

Rather than solving the equation analytically or through coupling with a full hydrodynamic model, this study implements a grid-based numerical approximation using finite differences. This approach balances physical interpretability with computational accessibility, allowing visualization of key dynamics such as lateral diffusion, wind-driven drift, and biodegradation decay. The oil spill drift simulations presented in this study are grounded in the classical advection–diffusion–reaction equation, which describes the transport and transformation of substances in a fluid medium. This physical process is typically expressed as:

$$\frac{\partial C}{\partial t} = D \nabla^2 C - \nabla \cdot (\vec{v} C) - \lambda C \quad (7)$$

where:

- $C(x, y, t)$ is the oil concentration at time t and location (x, y) ,
- D is the turbulent diffusion coefficient,
- $\vec{v}$ is the velocity field representing wind or ocean currents,
- λ is the combined rate of natural degradation and evaporation.

Rather than solving this equation analytically or using a full-scale hydrodynamic coupling approach, the present study adopts a grid-based numerical approximation using finite difference methods. This strategy offers a balance between physical interpretability and computational accessibility, enabling the visualization of key dynamics such as horizontal spreading, wind-driven transport, and degradation processes over time.

Each simulation step models the evolution of oil within a spatially discretized domain, where each grid cell represents an area of approximately 500×500 m, and each timestep corresponds to approximately 30 min of real-world evolution. These values can be adjusted to calibrate the model for finer or broader scales as needed. Physical constants—such as advection and diffusion rates—are selected in line with values commonly used in oil spill modeling studies, allowing the model to remain consistent with existing simulation benchmarks [27].

This simplified but scalable formulation underpins the simulation logic implemented in Section 4.1 (conceptual 2D model) and is further extended in Section 4.2 to incorporate vertical dispersion, real ocean current datasets, and satellite-based oil detection. As such, the framework serves both as a demonstrative educational tool and a flexible research platform for simulating pollutant behavior in marine and coastal environments.

4. Development of a Python Model for Oil Spill Drift Simulation

To analyse the dynamics of oil spills, a simplified computational model has been developed and implemented in Python using the equations defined in the chapter above. The goal of this model assumed by the authors was to simulate the movement, spreading, and degradation of an oil slick in a controlled environment as a ground stage toward a more complex algorithm. Starting from this type of simple model, a Python computer code was developed and implemented to simulate oil spill drift model, the algorithm being basically used to analyse the temporal evolution of oil slicks. At this preliminary stage, the authors' focus was on maintaining the code as minimalist as possible, to facilitate a clear and intuitive understanding of the underlying principles.

For oil spill drift simulation, the scalar field $f(t, x, y, z)$ has been defined as follows:

- it takes a value of '0' for water;
- it takes a value of '1' for the initial oil slick, meaning that the initial "thickness" of the oil layer is set to 1 (in dimensionless units);
- it can vary between 0 and 1, where an intermediate value (less than 1) indicates a reduction in the oil film's thickness due to diffusion and decomposition.

4.1. Problem Solution with Applied Code Lines in Python—Simple Model

For valuing the initial study hypothesis, the authors have developed a simple python code using the theoretical statements from the theoretical part for the advection, diffusion, and reaction study, as in Table 4 below.

Table 4. Simple Model—Python Code.

```

#No boundary conditions in this code (for simplicity, keeping the code minimal)
#Basically make the water grid large enough vs. oil spill such that the oil does not reach the boundary

# Import necessary libraries
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.widgets import Slider
from matplotlib.animation import FuncAnimation

# Parameters for simulation
SIZE_OF_GRID = 120
SIZE_OF_WIDTH_INITIAL_SPILL = 5
nn2 = int(SIZE_OF_GRID * 3/4)
INITIAL_OIL_CONCENTRATION = 1.0
IMG_FACTOR = 8.0 # Cut-off for oil color imaging

KRATE_diffusion = 0.1
KRATE_advection = 0.05
KRATE_evaporate = 0.001
KRATE_biodegradation = 0.0005

oil_concentration = INITIAL_OIL_CONCENTRATION
grid = np.zeros((SIZE_OF_GRID, SIZE_OF_GRID))

# Initial oil spill
nn2_i = max(0, nn2 - SIZE_OF_WIDTH_INITIAL_SPILL)
nn2_f = min(SIZE_OF_GRID, nn2 + SIZE_OF_WIDTH_INITIAL_SPILL)
for i in range(nn2_i, nn2_f + 6):
    for j in range(nn2_i, nn2_f + 1):
        grid[i, j] = oil_concentration

# Wind parameters
wind_speed = 1
wind_dir = (1, 1)

def get_derivatives(grid, KRATE_diffusion, KRATE_advection):
    grid_NEW = grid.copy()

    # Diffusion using numpy roll (more stable)
    diffusion = (
        np.roll(grid, 1, axis = 0) + np.roll(grid, -1, axis = 0) +
        np.roll(grid, 1, axis = 1) + np.roll(grid, -1, axis = 1) -
        4 * grid
    )
    grid_NEW += KRATE_diffusion * diffusion

    # Advection (mass-conserving using a temporary grid)
    new_grid = np.zeros_like(grid)
    for i in range(1, SIZE_OF_GRID - 1):
        for j in range(1, SIZE_OF_GRID - 1):
            new_i = i - wind_dir[1] * wind_speed
            new_j = j - wind_dir[0] * wind_speed

```

Table 4. *Cont.*

```

        if 0 <= new_i < SIZE_OF_GRID and 0 <= new_j < SIZE_OF_GRID:
            d_tr = grid[i, j] * KRATE_advection
            new_grid[new_i, new_j] += d_tr    # Add transported oil to target
            new_grid[i, j] -= d_tr           # Remove from source

    grid_NEW += new_grid

# Evaporation and biodegradation
    grid_NEW -= (KRATE_evaporate + KRATE_biodegradation) * grid_NEW

    return grid_NEW

# RK4 integrator
def integrator_step(grid, KRATE_diffusion, KRATE_advection):
    k1 = get_derivatives(grid, KRATE_diffusion, KRATE_advection)
    k2 = get_derivatives(grid + 0.5 * k1, KRATE_diffusion, KRATE_advection)
    k3 = get_derivatives(grid + 0.5 * k2, KRATE_diffusion, KRATE_advection)
    k4 = get_derivatives(grid + k3, KRATE_diffusion, KRATE_advection)

    grid_NEW = grid + (k1 + 2.0 * k2 + 2.0 * k3 + k4)/6.0

# Conservation law
    total_oil = np.sum(grid)
    total_new_oil = np.sum(grid_NEW)
    if total_new_oil > 0:
        grid_NEW = (grid_NEW/total_new_oil) * total_oil
    return grid_NEW

# Animation setup
fig, ax = plt.subplots()
plt.subplots_adjust(bottom = 0.25)
img = ax.imshow(grid, cmap = 'viridis', vmin = 0, vmax = oil_concentration/IMG_FACTOR)
ax.set_title("Oil Spill Simulation with RK4 Integrator")

# Sliders for parameters
axis_diffusion = plt.axes([0.1, 0.01, 0.65, 0.03])
slider_diffusion = Slider(axis_diffusion, 'KRATE diffusion', 0.0, 0.5, valinit = KRATE_diffusion)
axis_advection = plt.axes([0.1, 0.05, 0.65, 0.03])
slider_advection = Slider(axis_advection, 'KRATE advection', 0.0, 0.1, valinit = KRATE_advection)

def animate(frame):
    global grid
    KRATE_diffusion = slider_diffusion.val
    KRATE_advection = slider_advection.val
    grid = integrator_step(grid, KRATE_diffusion, KRATE_advection)
    img.set_array(grid)
    return img,

ani = FuncAnimation(fig, animate, frames = 100, interval = 100, blit = False)
plt.show()

```

The logic flow of the code developed in Table 4 is based on following programming flowchart:

1. import libraries, bringing in NumPy v.1.19.3 for math, Matplotlib v.3.10.1 for plots, and slider/animation tools;
2. define parameters by setting constants like grid size, oil concentration, and physical coefficients (diffusion, advection, etc.);
3. initialize Grid by creating a 2D water surface and places an initial oil patch, then assign oil concentration in center;
4. wind parameters definition to control oil drift direction as wind vector to define advection flow;
5. computes how oil spreads and moves by `get_derivatives()` for diffusion (Laplacian), advection (wind-shift) and reaction (evaporation & biodegradation);
6. uses Runge-Kutta 4th order to evolve the grid through time—`integrator_step()`—ensures smooth and stable simulation, that combines 4 stages: k_1 , k_2 , k_3 , k_4 ;
7. add mass conservation to adjust new grid so total oil stays constant (unless evaporated/degraded);
8. visualize initial Grid, using `imshow()` to display oil slick as a coloured image;
9. add `Slider()` to allow real-time user adjustment of advection & diffusion;
10. updates grid for each animation frame using `integrator` using `animate()` function;
11. launch animation and show the simulation in motion by `FuncAnimation()`.

An example of an oil spill evolution simulation applying the above code from Table 4 is shown in Figure 2, developed by the authors using the parameters for the constants of advection, diffusion, reaction (reaction = evaporation + biodegradation). In this representation, the initial oil slick is depicted in yellow, while the seawater is represented in blue; the white arrow indicates the direction of advection. The simulation highlights the following physical processes:

- advection-driven displacement—the center of mass of the oil slick moves in the direction of the advection flow, as expected based on physical principles;
- diminishing field values over time—the scalar field $f(t,x,y,z)$ gradually decreases, corresponding to a reduction in the yellow coloration in the graphical representation—this is due to the combined effects of evaporation, degradation, and diffusion;
- lateral spreading—the oil slick expands over a larger surface area as a result of diffusion.

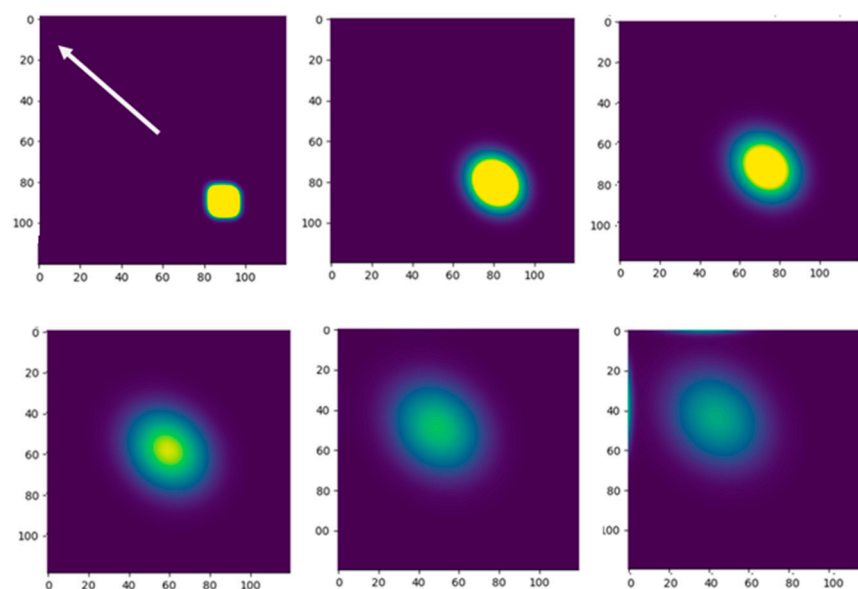


Figure 2. Time evolution of oil slick, modelled with advection, diffusion, and reaction in the open ocean. Source: authors' developments applying Python code.

From this preliminary modelling stage, the following conclusions can be drawn:

1. the simulation provides insight into the basic physical mechanisms responsible for oil spill motion and transformation, including advection, diffusion, and chemical reactions, confirming the fundamental physical laws governing oil spill dynamics;
2. the model demonstrates how these fundamental laws can be incorporated into a basic Python algorithm testing the implementation of these principles in a simplified computational model;
3. despite its simplicity, the implemented code validates the expected physical behaviour successfully capturing the expected physical behaviour of oil spill evolution, reproducing the key phenomena of drift, dispersion, and degradation.

This foundational step serves as a basis for developing more complex simulations, incorporating additional real-world factors such as wind influence, variable ocean currents, and chemical interactions affecting the oil's behaviour over time.

4.2. Study Proposal for an Enhanced Model—Python Algorithm for Enriched Modelling

The code from Table 4 is a starting point for many real-world simulations with solid consistency for further practical applications, with some modifications having the potential to become a valuable tool for scientists, engineers, and policymakers across different industries. To improve the initial version of the code presented in Table 4, the authors have compiled the suggestions drawn from the literature review and case studies analysed and described in first chapters, as following:

- adding real-world ocean currents data, to make the simulation more accurate—the NOAA Global Ocean Currents dataset may provide real-time current data to be included in the model, making the proposed code more dynamic, incorporating real-world ocean physics;
- implementing remote sensing for autonomous tracking—adding drone or satellite-based oil detection may enable real-time spill tracking, allowing the model to detect oil spills from images and to incorporate it into your simulation grid;
- simulating subsurface oil dispersion—as some oil spills sink below the surface and behave differently due to pressure and density, the model may need vertical diffusion modelling, to simulate both surface and underwater oil spill dynamics;
- expanding to inland water bodies—some spills occur in rivers and lakes, which behave differently as the inland water movement is governed by river flow, not ocean currents and this may make the model applicable also to river and lake environments;
- including chemical pollution transport, as the industrial spills involve chemicals, not just oil—as chemicals dissolve in water differently based on solubility that may improve the model by adding chemical solubility and degradation effects, that the code to simulate both oil and chemical spills.

The final improved version of the oil spill simulation software in Python 3.13.0, will integrate the real-world ocean currents (NOAA Dataset), remote sensing—satellite-Based Spill Detection (OpenCV Image Processing), subsurface oil dispersion (3D Diffusion), inland flow modelling (River Flow Advection), and chemical transport and biodegradation, resulting the improved code sequence under Python depicted in Table 5.

Table 5. Complex Simulation Model—Enriched Python Code.

```

import numpy as np
import xarray as xr
import cv2
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation

# Simulation Parameters
GRID_SIZE = 120 # Grid resolution
DEPTH_LAYERS = 10 # Layers for subsurface dispersion
INITIAL_OIL_CONCENTRATION = 1.0
IMG_FACTOR = 8.0

# Diffusion & Advection Rates
KRATE_DIFFUSION = 0.1
KRATE_ADVECTION = 0.05
KRATE_VERTICAL_DIFFUSION = 0.02
BIODEGRADATION_RATE = 0.01

# Load Real-World Ocean Currents (NOAA Dataset)
dataset_url = "https://data.nodc.noaa.gov/thredds/dodsC/modeldata.nc" (accessed on 1 January 2025).
ds = xr.open_dataset(dataset_url)

u_current = ds['u'].isel(time = 0).values
v_current = ds['v'].isel(time = 0).values

# Resize to match simulation grid
u_current = np.interp(np.linspace(0, 1, GRID_SIZE), np.linspace(0, 1, u_current.shape[0]), u_current)
v_current = np.interp(np.linspace(0, 1, GRID_SIZE), np.linspace(0, 1, v_current.shape[1]), v_current)

# Initialize Grid for Oil Spill
grid = np.zeros((DEPTH_LAYERS, GRID_SIZE, GRID_SIZE))
grid[0, GRID_SIZE//2-3:GRID_SIZE//2+3, GRID_SIZE//2-3:GRID_SIZE//2+3] =
INITIAL_OIL_CONCENTRATION

# Remote Sensing (Oil Spill Detection from Image)
def detect_oil_spill(image_path):
    img = cv2.imread(image_path)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    _, thresh = cv2.threshold(gray, 100, 255, cv2.THRESH_BINARY)
    contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

    oil_spill_grid = np.zeros((GRID_SIZE, GRID_SIZE))
    for contour in contours:
        cv2.drawContours(oil_spill_grid, [contour], -1, 255, thickness = -1)

    return (oil_spill_grid/255) * INITIAL_OIL_CONCENTRATION

# Diffusion & Advection Model
def get_derivatives(grid):
    grid_new = grid.copy()

```

Table 5. Cont.

```

# Horizontal Diffusion (Surface & Subsurface)
horizontal_diffusion = (
    np.roll(grid, 1, axis = 1) + np.roll(grid, -1, axis=1) +
    np.roll(grid, 1, axis = 2) + np.roll(grid, -1, axis=2) - 4 * grid
)
grid_new += KRATE_DIFFUSION * horizontal_diffusion

# Vertical Diffusion (Subsurface Oil Spread)
vertical_diffusion = (
    np.roll(grid, 1, axis = 0) + np.roll(grid, -1, axis = 0) - 2 * grid
)
grid_new += KRATE_VERTICAL_DIFFUSION * vertical_diffusion

# Advection (Improved Upwind Scheme—Mass-Conserving, Direction-Aware)
# This version uses upwind differencing based on current direction (u, v)
# It ensures realistic oil transport and mass conservation across the grid
for i in range(1, GRID_SIZE - 1):
    for j in range(1, GRID_SIZE - 1):
        u = u_current[i, j]
        v = v_current[i, j]
        if u > 0:
            flux_x = u * (grid[0, i - 1, j] - grid[0, i, j])
        else:
            flux_x = u * (grid[0, i, j] - grid[0, i + 1, j])
        if v > 0:
            flux_y = v * (grid[0, i, j - 1] - grid[0, i, j])
        else:
            flux_y = v * (grid[0, i, j] - grid[0, i, j + 1])
        advection_flux = KRATE_ADVECTION * (flux_x + flux_y)
        grid_new[0, i, j] += advection_flux

# River Flow Advection (For Inland Water Spills)
river_flow_speed = 0.05
river_direction = (1, 0) # Flowing in x-direction

for i in range(1, GRID_SIZE - 1):
    for j in range(1, GRID_SIZE - 1):
        new_i = i + int(river_direction[0] * river_flow_speed)
        new_j = j + int(river_direction[1] * river_flow_speed)
        if 0 <= new_i < GRID_SIZE and 0 <= new_j < GRID_SIZE:
            grid_new[0, new_i, new_j] += grid[0, i, j] * KRATE_ADVECTION
            grid_new[0, i, j] -= grid[0, i, j] * KRATE_ADVECTION

# Biodegradation (Oil Breakdown)
grid_new *= (1 - BIODEGRADATION_RATE)

return grid_new

# RK4 Integration
def rk4_step(grid):
    k1 = get_derivatives(grid)
    k2 = get_derivatives(grid + 0.5 * k1)
    k3 = get_derivatives(grid + 0.5 * k2)
    k4 = get_derivatives(grid + k3)
    return grid + (k1 + 2 * k2 + 2 * k3 + k4)/6.0

```

Table 5. Cont.

```

# Visualization
fig, ax = plt.subplots()
plt.subplots_adjust(bottom=0.25)
img = ax.imshow(grid[0], cmap = 'viridis', vmin = 0, vmax = INITIAL_OIL_CONCENTRATION/IMG_FACTOR)
ax.set_title("Oil Spill Simulation with Enhanced Features")

def animate(frame):
    global grid
    grid = rk4_step(grid)
    img.set_array(grid[0])
    return img,

ani = FuncAnimation(fig, animate, frames = 100, interval = 100, blit = False)
plt.show()

```

The logic flow of the code developed in Table 5 is based on following programming flowchart:

1. import libraries and load packages for computation, visualization, remote sensing, and ocean data;
2. set parameters by defining simulation resolution, rate constants, and initial values;
3. load ocean currents by bringing in and interpolating NOAA datasets to simulate real advection;
4. initialize 3D Grid by creating 3D matrix to track surface and subsurface oil;
5. remote sensing detection by converting images to oil grid using OpenCV for contour extraction;
6. get_derivatives() to calculate horizontal and vertical diffusion, advection, river flow, and decay;
7. apply Runge-Kutta method for grid evolution by RK4 Integration;
8. visualization to show surface oil concentration using color plots;
9. animate to define update function to simulate oil drift over time;
10. run simulation and animate using FuncAnimation to visualize results dynamically.

Comparing the advanced oil spill simulation model with the simplified version, several significant enhancements emerge that substantiate its increased realism, applicability, and scalability. While the basic model operates on a two-dimensional grid, the enriched version incorporates a three-dimensional structure, allowing for the simulation of both surface and subsurface oil dispersion. A key improvement lies in the integration of real-world ocean current data sourced from NOAA, replacing the fixed wind vector used in the initial version. This addition enables the simulation to reflect time-varying and location-specific hydrodynamic conditions with higher fidelity. Moreover, the advanced model supports the simulation of riverine or inland oil spills through custom advection logic, making it applicable not only to marine environments but also to lakes and rivers. Satellite and drone image-based oil detection is also introduced through OpenCV-based image processing, enabling the automated extraction of initial spill locations from remote sensing data. Biodegradation is treated more dynamically across depth layers, enhancing the accuracy of the chemical weathering process representation. Collectively, these improvements transform the simulation from a conceptual prototype into a modular, high-resolution modeling framework adaptable for emergency response, environmental planning, and cross-disciplinary pollutant dispersion studies.

4.3. Models' Limitations

Both modeling approaches presented in this study—outlined in Tables 4 and 5 offer distinct contributions toward understanding and simulating oil spill drift, but each also entails specific methodological limitations, being experimental prototypes. The solution in Table 4, although effective for visualizing fundamental transport processes such as advection, diffusion, and decay, operates within a highly idealized environment. It relies entirely on synthetic inputs, fixed wind conditions, and uniform parameter settings without grounding in real-world hydrodynamic or meteorological variability. As such, in this initial format, it lacks empirical calibration and is not intended for predictive use beyond conceptual demonstration or educational scenarios.

The enhanced model described in Table 5 represents a significant advancement by integrating real-time oceanographic datasets (e.g., NOAA currents), vertical diffusion structures, and remote sensing inputs. However, despite its increased realism and extensibility, this model has not yet been tested against observed oil spill events or validated through quantitative comparisons with field data, being developed so far as a proposed solution, to be further tested. Moreover, the computational performance under real-time conditions, the robustness of satellite spill detection algorithms, and the treatment of complex coastal geometries remain to be fully evaluated in future case studies conducted by the authors.

Future research will focus on the empirical validation of the enhanced model using historical spill data and on establishing partnerships with marine and environmental agencies for in situ testing. Sensitivity analysis, uncertainty quantification, and integration with ecological vulnerability layers will also be pursued to improve operational readiness and scientific reliability. These efforts are essential for transitioning the models from experimental prototypes to robust tools in marine risk mitigation and decision support systems.

4.4. Model Validation Strategy

Although the modeling results presented in this study demonstrate internal consistency and behavior aligned with theoretical expectations, formal validation against real-world data remains an essential step toward operational reliability. The current phase of development focuses on conceptual design and simulation coherence; however, future efforts will be directed at quantitative validation and calibration of the model through multiple complementary methods, as following:

- a. Satellite-based validation—one of the primary approaches will involve the comparison of simulated spill trajectories with actual oil slicks detected via synthetic aperture radar (SAR) or multispectral satellite imagery. Datasets from platforms such as Sentinel-1, RADARSAT, and MODIS-Aqua will be used to extract spatial patterns and temporal evolution of observed oil spills. These will then be overlaid with model outputs using geospatial tools (e.g., QGIS, Python's rasterio/geopandas) to assess spatial alignment, drift direction accuracy, and spread rates.
- b. Historical case replication—the model will also be applied retrospectively to documented spill events—such as the Deepwater Horizon incident or localized spills in the Black Sea or Mediterranean, where detailed oceanographic and wind data are available. Using corresponding NOAA or Copernicus marine datasets, the model's outputs will be compared to published drift trajectories and satellite-observed oil extents. Metrics such as centroid displacement error, area overlap index, and drift angle deviation will be employed to quantify performance.
- c. In-situ buoy and sensor data—where available, drifting buoys (e.g., NOAA SVP drifters) or sensor-equipped autonomous surface vehicles (ASVs/USVs) will be used as physical proxies to validate oil drift patterns. These objects simulate surface-layer

transport under wind and current conditions and can be directly compared with simulated particle trajectories.

- d. Statistical evaluation metrics—model performance will be assessed using standard validation metrics such as: Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) of centroid position, Intersection over Union (IoU) for slick area match, Skill Score (SS) comparing forecast vs. observed drift paths.
- e. Visual and behavioral corroboration—in addition to numerical validation, the model's ability to reproduce characteristic oil spill behaviors, such as elongated spread downwind, coastal accumulation, and decay over time, may be evaluated qualitatively using expert interpretation of imagery and documented incident reports.

5. Results Interpretations and Model Applications

The results derived from the simulation models provide valuable insights into the physical behaviour and evolution of oil spills in marine and coastal environments. In the initial simulations (Table 4), the simplified Python model seek to demonstrate core transport phenomena, namely advection, diffusion, and degradation. The progression of the oil slick, from a compact release into a diffused and partially degraded plume, aligns with the expected dynamics of marine hydrocarbon dispersion, validating the physical basis of the model. The simulation highlighted three key behaviours:

- advection-driven displacement, whereby the oil slick consistently moved in the prescribed wind direction, reflecting wind-driven surface transport;
- lateral diffusion, where the slick expanded in size while decreasing in concentration due to turbulent mixing;
- progressive decay of concentration values, caused by evaporation and biodegradation processes embedded in the scalar field evolution.

These outputs validate the fundamental structure of the model and reinforce its capability to simulate time-evolving oil distribution using simplified physics. However, the absence of real-world inputs (e.g., ocean currents, bathymetry) limits its predictive power for specific geographic regions. In contrast, the enhanced model introduced in the latter part of the study provided substantial improvements in simulation realism. The integration of NOAA ocean current datasets allowed the model to reflect regionally specific circulation patterns, offering a more accurate trajectory prediction. Additionally, the incorporation of a 3D grid structure enabled subsurface oil transport simulations, a critical feature for modelling deepwater blowouts or denser hydrocarbon fractions. This was particularly relevant when simulating scenarios where vertical mixing and sedimentation play significant roles in environmental impact. The enhanced model also introduced a novel capability for remote sensing data assimilation. By extracting spill contours from satellite or drone imagery, the model supports autonomous detection and real-time initialization, improving its operational relevance. Combined with river flow advection features and vertical degradation, the model demonstrated multi-scenario flexibility, making it applicable to inland water bodies, estuarine systems, and offshore zones.

The comparative performance between the two model versions underscores the importance of integrating environmental variability, data assimilation, and modular architecture. The enhanced model not only improved spatial and temporal resolution but also broadened the range of applicable scenarios, from research simulations to real-world contingency planning. The simulation results can support decision-making for emergency response, pollution impact assessment, and climate resilience strategies, especially when integrated with remote sensing and ecological sensitivity layers.

The model designed in Python code, may be applied for different scenarios in multi-purpose approach, reaching wide version of further implementation, depicted as following:

- a. *Environmental modelling for oil spill tracking and prediction*—the presented model simulation can help predict the spread of oil spills in oceans and rivers and can be integrated with real-world data (e.g., wind speed, ocean currents, and temperature). Environmental agencies (e.g., NOAA, EPA) can use this model to optimize oil spill cleanup strategies. Also, this model can be used in care of predicting the movement of an oil spill near a coastline to allocate response teams effectively.
- b. *Atmospheric and oceanic pollution dispersion*—the diffusion and advection principles applied in the enriched code can be adapted to model air pollution spread, valuable to predict how smog, wildfire smoke, or chemical leaks may disperse over time. The model can be practically used for forecasting the spread of toxic gas leaks from industrial sites to warn nearby populations.
- c. *Epidemic and disease spread modelling*—similarly to how oil disperses, the disease spread (e.g., COVID-19, Ebola) following diffusion and advection principles, then the model may be extended to simulate disease transmission in a population over time. As application, the model can be in this case used to simulate how infectious diseases spread in a city based on population movement.
- d. *Industrial and chemical spill simulation*—this model can be adapted to track toxic chemical leaks in rivers, lakes, and industrial zones, providing a useful tool for pre-preparation to simulate potential spills from chemical plants in case of accidents scenarios. As example, the model can be implemented to predict the flow of a chlorine gas leak from a factory to nearby communities.
- e. *Geophysical and climate research*—the model can be used to model sediment transport in rivers and oceans, applicable to simulate lava flow, glacier movements, and sand dune formation.
- f. *Traffic and crowd flow simulation*—by adapting advection-diffusion principles, the code can model the traffic congestion in cities, the pedestrian flow in crowded areas, evacuation scenarios for population in crisis scenarios.
- g. *Space science and astrophysics*—the diffusion model can be applied to simulate interstellar gas diffusion in space or the atmospheric dispersion of gases on other planets.
- h. *Material science and nanotechnology*—the mode can be used to simulate liquid diffusion in porous materials, for drug diffusion in biological tissues or other particular applications.

Building upon the results of this study, the developed simulation models offer a valuable foundation for designing operational systems that mitigate environmental risks associated with oil spills and hydrocarbon pollution in marine ecosystems. Integrating these models with modern sensor platforms and AI-enhanced monitoring tools can significantly improve the resilience and responsiveness of coastal and offshore environmental management.

One key application of the enhanced model lies in early warning systems that combine satellite data with predictive modelling to forecast spill trajectories and prioritize areas of intervention [28]. For instance, integrating real-time ocean current data with an advection-diffusion simulation enables accurate forecasting of the spatial and temporal evolution of an oil slick, guiding the placement of containment booms, dispersants, and response vessels.

Additionally, riverine, and coastal oil spill scenarios, traditionally underserved in marine modelling, can be simulated using customized advection logic, as demonstrated in this study. This is critical in estuarine zones and industrial ports where oil spill impact can extend rapidly to freshwater systems being emphasized that rapid detection and targeted response significantly reduce ecological damage in such high-risk areas [29].

Modern marine monitoring frameworks now increasingly rely on multi-source remote sensing systems supported by artificial intelligence. Satellites such as Sentinel-1

and RADARSAT provide large-scale Synthetic Aperture Radar (SAR) coverage, while Unmanned Aerial Vehicles (UAVs) supply higher-resolution imagery at tactical levels [30,31]. These visual inputs, when processed through deep learning algorithms, can automatically detect and classify oil slicks [32]. UAV platforms can further be integrated with autonomous surface vehicles and sensor buoys to measure hydrocarbon concentrations, turbidity, and salinity, providing continuous feedback loops to the simulation engine [33].

The designed code may be used for cloud-based platforms allowing the real-time data assimilation, user-friendly dashboards, and automated alerts. These tools collectively contribute to a Dynamic Decision Support System (DSS), which can be critical in coordinating response between governmental, industrial, and civil actors [34].

The integration of simulation-based prediction using similar intelligent codes as the present developed ones, may offer tangible benefits for monitoring and environment awareness as following:

- faster incident response times, reducing spill spread and damage;
- prioritized protection of sensitive areas such as fisheries, coral reefs, and mangroves;
- data-driven planning of oil transport routes and coastal infrastructure;
- support for compliance with MARPOL, UNCLOS, and regional marine protection treaties;
- these tools may empower marine and coastal management agencies with predictive capabilities rather than reactive containment, transforming environmental risk mitigation into a proactive, real-time process.

Although, while several open-source modelling platforms have been developed to address the simulation of oil spill drift in marine environments, the proposed Python-based model introduced in this study presents a distinct and valuable contribution. Open frameworks such as GNOME developed by NOAA, OpenDrift by the Norwegian Meteorological Institute [35], and Parcels (Probably A Really Computationally Efficient Lagrangian Simulator) [36] are widely recognized for their robust capabilities in Lagrangian particle tracking, operational spill forecasting, and integration with hydrodynamic datasets. These tools, while powerful, are largely optimized for large-scale operational use, with modular but complex architectures requiring multiple dependencies, and they often emphasize individual particle trajectories rather than grid-based field evolution. In contrast, the present study proposes a custom-built Eulerian approach based on advection-diffusion-reaction partial differential equations (PDEs), which enhances interpretability and didactic utility while enabling rapid prototyping and adaptation across diverse environmental scenarios. Moreover, unlike GNOME or OpenDrift, the developed model integrates a novel remote sensing pre-processing module utilizing OpenCV for satellite or UAV-based oil spill detection. This functionality allows for automated initialization of oil spill positions directly from imagery, a capability not natively included in the aforementioned platforms.

In terms of environmental realism, the model incorporates 3D vertical diffusion, biodegradation kinetics, and dynamic current data from NOAA, bridging the gap between simplified academic models and operational simulation tools. Its lightweight Python architecture with minimal external dependencies makes it especially suitable for educational use, research training, and interdisciplinary applications, including the simulation of drifting sea mines, inland chemical spills, and even diffusion-based scenarios in atmospheric or biological systems. Therefore, the present developed code provides a complementary alternative to existing open-source tools, combining conceptual transparency, cross-domain flexibility, and real-world integration, offering a scalable pathway toward decision support systems and adaptive environmental modelling.

6. Conclusions

This study presents significant advancements in oil spill drift modelling, emphasizing the integration of hydrodynamic simulations, artificial intelligence, and remote sensing technologies to enhance predictive accuracy and response efficiency. The research has aimed to provide a comprehensive review of state-of-the-art methodologies while introducing novel modelling frameworks that contribute to the evolution of marine pollution management strategies.

The study offers a multi-faceted approach to oil spill drift prediction by combining Lagrangian particle tracking, Eulerian grid-based modelling, and AI-driven techniques. The inclusion of stochastic simulations and real-time satellite data assimilation has aimed to improve the forecast accuracy, making this research particularly valuable for crisis response teams and environmental monitoring agencies. Moreover, the article systematically integrates Stokes drift effects and ecological sensitivity models, filling a critical gap in existing oil spill trajectory simulations. The authors' contributions to the state-of-the-art of modelling of oil spill drift, can be concluded in the following outlines:

- introduction of hybrid models—the study integrates Deepwater Oil Spill Model (DWOSM) with stochastic and machine learning algorithms, improving the parameterization of wind-driven drift;
- AI-Driven forecasting—the authors employ adversarial neural networks and support vector regression (SVR) to refine oil spill trajectory simulations, significantly reducing model uncertainty;
- empirical validation—the proposed models may be successfully tested in real-world case studies, demonstrating their effectiveness in emergency response planning and coastal protection [27];
- real-time data integration—the research advances satellite-assisted oil spill detection, enhancing the real-time operational capabilities of existing forecasting models.

The enhanced oil spill drift models developed in this study may have broad applications across multiple sectors, including:

- marine environmental protection—improved predictive modelling aids in mitigating the impact of oil spills on marine ecosystems;
- emergency response and crisis management—real-time spill tracking enhances decision-making for rapid response efforts, using modelling software as GNOME v.47, ADIOS v.2.0.12 or OpenDrift v.1.0;
- coastal and offshore infrastructure protection—by forecasting oil drift patterns allows ports, fisheries, and energy sectors to implement proactive risk mitigation strategies;
- regulatory compliance and policy making—the integration of ecological sensitivity modelling informs environmental regulations and sustainable maritime policies.

To further advance oil spill drift modelling, the authors' are foreseeing future research should focus on: multi-source data fusion (by combining satellite, UAVs and radar observations for enhanced real-time spill detection), high-resolution computational modelling (expanding current models to simulate complex subsurface oil spills and sediment transport interactions), AI-enhanced predictive analytics (developing deep learning algorithms capable of learning from historical oil spill events for continuous model improvement) and operationalization for crisis management (enhancing computational efficiency to deploy AI-driven oil spill simulations in real-time response scenarios).

This research bridges the gap between theoretical oil spill modelling and practical emergency response applications, offering a cutting-edge framework that significantly enhances prediction accuracy, crisis response, and environmental risk assessment. By integrating computational intelligence, remote sensing, and machine learning, the study

sought to set a new benchmark in oil spill drift modelling, paving the way for future innovations in marine pollution management and climate resilience strategies.

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